

# Chapter 3, Section 11

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1. Show that the result of (11.2) is false without the projective hypothesis. For example, let  $X = \mathbb{A}_k^n$ , let  $P = (0, \dots, 0)$ , let  $U = X - P$ , and let  $f : U \rightarrow X$  be the inclusion. Then the fibers of  $f$  all have dimension 0, but  $R^{n-1} f_* \mathcal{O}_U \neq 0$ .

*Proof.* By (8.2) and (8.5), we have the isomorphism

$$R^{n-1} f_* \mathcal{O}_U|_U \cong \widehat{H^{n-1}(U, \mathcal{O}_U)},$$

so it suffices to show  $H^{n-1}(U, \mathcal{O}_U) \neq 0$ . This is a generalization of (Ex. 4.3), and we compute  $H^{n-1}(U, \mathcal{O}_U)$  via Čech cohomology. Let  $x_1, \dots, x_n$  be affine coordinates of  $X$ , i.e.,  $X = \text{Spec } k[x_1, \dots, x_n]$ , and take  $\mathcal{U}$  to be the open covering of  $U$  by the open sets  $U_i = X - \{x_i \neq 0\}$ . Setting  $V = \bigcap_{i=1}^n U_i$  and  $V_j = \bigcap_{i \neq j} U_i$ , the last two terms of the Čech complex are

$$C^{n-2} = \prod_{j=1}^n \Gamma(V_j, \mathcal{O}_U), \quad C^{n-1} = \Gamma(V, \mathcal{O}_U).$$

Now

$$\begin{aligned} \Gamma(V_j, \Omega) &= k \left[ x_1, \dots, x_n, \frac{1}{x_1}, \dots, \frac{1}{x_j}, \dots, \frac{1}{x_n} \right], \\ \Gamma(V, \Omega) &= k \left[ x_1, \dots, x_n, \frac{1}{x_1}, \dots, \frac{1}{x_n} \right], \end{aligned}$$

and the image of the boundary map  $d : C^{n-2} \rightarrow C^{n-1}$  consists of all expressions  $x_1^{r_1} \cdots x_n^{r_n}$  where at least one of  $r_1, \dots, r_n$  is non-negative. Hence,  $H^{n-1}(U, \mathcal{O}_U)$  consists of all expressions  $x_1^{r_1} \cdots x_n^{r_n}$  with  $r_i < 0$  for all  $i$ .  $\square$

2. Show that a projective morphism with finite fibers is a finite morphism.

*Proof.* Let  $f : X \rightarrow Y$  be a projective morphism of Noetherian schemes, and let  $Y' = \text{Spec } f_* \mathcal{O}_X$ . The natural map  $g : Y' \rightarrow Y$  is finite, and we get a projective morphism  $f' : X \rightarrow Y'$  with connected fibres such that  $f$  factors into  $g \circ f'$  (11.5). The property of quasi-finiteness satisfies the conditions of (II, Ex. 4.8), and finite morphisms are separated, so  $f'$  is quasi-finite. The composition of finite morphisms is finite, so it suffices to show  $f'$  is a finite morphism. Thus, we reduce to showing projective morphisms with finite and connected fibres is a finite morphism. By (II, Ex. 3.4), we further reduce to the case  $Y = \text{Spec } A$  for some Noetherian ring  $A$  and show  $X = \text{Spec } B$ , where  $B$  is a finite  $A$ -algebra. If  $f$  has finite and connected fibres, then  $f$  is injective as a map between topological spaces. Since projective morphisms are closed mappings,  $f$  is a homeomorphism of  $X$  onto a closed subscheme of  $Y$ . The result follows from (II, Ex. 3.11).  $\square$

8. Let  $f : X \rightarrow Y$  be a projective morphism, let  $\mathcal{F}$  be a coherent sheaf on  $X$  which is flat over  $Y$ , and assume that  $H^i(X_y, \mathcal{F}_y) = 0$  for some  $i$  and some  $y \in Y$ . Then show that  $R^i f_*(\mathcal{F})$  is 0 in a neighborhood of  $y$ .

*Proof.* Since  $R^i f_*(\mathcal{F})$  is a coherent sheaf on  $Y$  (8.8b), it suffices to show  $R^i f_*(\mathcal{F})_y = 0$ . The question is local on  $Y$ , so we reduce to the case  $Y$  is affine, equal to the spectrum of the local Noetherian ring  $A = \mathcal{O}_y$ , and  $y$  is the closed point of  $Y$ . Let  $\hat{A}$  denote the completion of  $A$  with respect to its maximal ideal  $\mathfrak{m}$ . Consider the natural isomorphism of (11.1)

$$\widehat{R^i f_*(\mathcal{F})}_y \cong \varprojlim H^i(X_n, \mathcal{F}_n).$$

We claim it suffices to show the right hand side is zero. Indeed, by (8.2) and (8.5), we have the isomorphism

$$R^i f_*(\mathcal{F})_y \cong H^i(X, \mathcal{F}),$$

and according to (A.M., 10.13), we also have the isomorphism

$$\widehat{R^i f_*(\mathcal{F})}_y \cong H^i(X, \mathcal{F}) \otimes \hat{A},$$

so the claim follows from the fact that  $\hat{A}$  is a faithfully flat  $A$ -algebra.

Now, we will show the natural map  $H^i(X_{n+1}) \rightarrow H^i(X_n, \mathcal{F}_n)$  is injective for every  $n > 1$ . Admitting this for the moment, since  $X_1 = X_y$  and  $\mathcal{F}_1 = \mathcal{F}_y$ , it follows by induction that

$$H^i(X_n, \mathcal{F}_n) \hookrightarrow H^i(X_y, \mathcal{F}_y) = 0,$$

which in turn implies the inverse limit is zero, which is what we want.

Consider the exact sequence of  $A$ -modules

$$0 \longrightarrow \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow A/\mathfrak{m}^{n+1} \longrightarrow A/\mathfrak{m}^n \longrightarrow 0.$$

Since  $\mathcal{F}$  is flat over  $Y$ , tensoring with  $\mathcal{F}$  is an exact functor, so we have an exact sequence

$$0 \longrightarrow \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow \mathcal{F}_{n+1} \longrightarrow \mathcal{F}_n \longrightarrow 0.$$

Taking the long exact sequence in cohomology gives

$$H^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) \longrightarrow H^i(X_{n+1}, \mathcal{F}_{n+1}) \longrightarrow H^i(X_n, \mathcal{F}_n),$$

so it suffices to show the term on the left is zero. Since  $\mathfrak{m}^n/\mathfrak{m}^{n+1}$  is a finite-dimensional vector space over  $k(y)$ , and cohomology commutes with direct sums, we have the isomorphism

$$H^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) \cong H^i(X, \mathcal{F} \otimes k(y)) \otimes_{k(y)} \mathfrak{m}^n/\mathfrak{m}^{n+1},$$

where  $k(y)$  is the constant sheaf  $k(y)$  on the closed point  $y$  of  $Y$ . Thus, we can apply the isomorphism of (9.4)

$$H^i(X, \mathcal{F} \otimes k(y)) \cong H^i(X_y, \mathcal{F}_y) = 0$$

to conclude that  $H^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) = 0$ .

□